Fiscal Year:	FY 2019	**Task Last Updated:**	FY 08/12/2020
PI Name:	Globus, Ruth Ph.D.		
Project Title:	Free Radical Theory of Aging in Space		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology (2) Neurobiology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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PI Organization Type:	NASA CENTER	**Phone:**	650-604-5247
Organization Name:	NASA Ames Research Center		
PI Address 1:	Bone and Signaling Laboratory		
PI Address 2:	Space Biosciences Research Branch		
PI Web Page:			
City:	Moffett Field	**State:**	CA
Zip Code:	94035-1000	**Congressional District:**	18
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	08/01/2016	**End Date:**	05/31/2020
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 5/31/2020 per F. Hernandez/ARC (Ed., 8/12/2020) NOTE: End date changed to 7/31/2020 per F. Hernandez/ARC (Ed., 4/13/18) NOTE: Change in period of performance to 8/1/2016-7/31/2018 (originally 11/1/2014-10/31/2017) per F. Hernandez/ARC (Ed., 7/29/16)		
Key Personnel Changes/Previous PI:	October 2017 report: New Co-Investigators: Candice Tahimic (NASA Ames Research Center); David Goukassian (Temple University). Remove Co-Investigator: Mohammed Shahnazari.		
COI Name (Institution):	Alwood, Joshua Ph.D. (NASA Ames Research Center) Berkowitz, Daniel M.D., Ph.D. (Johns Hopkins School of Medicine) Limoli, Charles Ph.D. (University of California, Irvine) Santhanam, Lakshmi Ph.D. (Johns Hopkins University) Schreurs, Ann-Sofie Ph.D. (NASA Ames Research Center) Tahimic, Candice Ph.D. (NASA Ames Research Center) Goukassian, David M.D., Ph.D. (Temple University)		
Grant/Contract No.:	Internal Project		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	Many of the adverse physiological effects of spaceflight on astronauts, such as cardiovascular and musculoskeletal deconditioning, resemble those of aging on Earth, leading to the hypothesis that reduced gravity accelerates human aging. This idea is based on numerous spaceflight findings at the organ-system level observed in both humans and animals. Possible specific, mechanistic links between aging and gravity at the molecular level have yet to be established; this is the knowledge gap that we will address in the proposed research project. Our Specific Aims will test responses to long duration spaceflight of a range of tissues (i.e., neural, cardiac, skeletal) in both wild type (control) and transgenic mice. The transgenic mice are engineered to have less oxidative damage in mitochondria, which may occur as a consequence of exposure to long duration spaceflight as well as progressive aging. Female mice (both wildtype and transgenic) will be exposed to spaceflight or kept as ground controls for as long as two months, then tissues (neural, heart, bone) recovered for subsequent analysis of gene expression, oxidative stress biomarkers, and tissue functions. Thus, results from this research will identify basic molecular mechanisms that may underlie degenerative changes arising during prolonged spaceflight or progressive aging on Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Results from this research will identify basic molecular mechanisms that may underlie degenerative changes arising during prolonged spaceflight or progressive aging on Earth.
Task Progress:	NOTE: Compiled from report provided to NASA Ames in 2019; received by Task Book May 2020. The ground-based phase of the project focused on testing of our central hypothesis as described in Aim 1: Determine if spaceflight causes oxidative damage to tissues (cardiovascular system, nervous system, bone) and accelerates tissue degeneration relative to ground controls. We used hindlimb unloading (HU) to simulated weightlessness over a 30 day period, with both wildtype (WT) and transgenic mice engineered to overexpress the human catalase gene in the mitochondria (mCAT). To more closely mimic the hardware intended to house our flight mice in space (Rodent Research Project-RRP), we developed a novel method to hindlimb unload the mice in pairs. This was deemed important since RRP's mouse Habitat houses the mice in groups and social isolation can profoundly affect biological responses to various stimuli. HU experiments are typically done in social isolation. We compared responses of WT and mCAT mice over a 30 day HU period in 16-week old female mice, in either single or social housing. HU in both single and social housing conditions led to expected decrements in bone (tibia) microstructure and soleus weight compared to controls, suggesting that these changes are primarily responses to altered gravitational loading (i.e., hindquarter disuse and fluid shifts) (Tahimic et al., 2019--see Bibliography). In contrast, some (not all) responses of the immune system (Paul et al., in review) and hypothalamic-pituitary-adrenal (HPA) axis responses were differentially impacted by the social environment during HU (Tahimic et al., 2019--see Bibliography). We also developed a microcomputed tomography method to evaluate heart structure and found that mCAT mice had greater left ventricular wall thickness. There was a trend for an HU effect to reduce left ventricular cavity volume. No obvious differences due to genotype were evident in the responses to HU; hence in the context of structure, this finding resembles our findings from muscle and bone. Mechanical testing of aorta has been conducted and analysis is ongoing. In contrast to the musculoskeletal system and heart tissue, in the brain, simulated weightlessness and social isolation each caused changes in levels of inflammatory cytokines in the hippocampus. Interestingly, the repertoire of cytokines regulated by HU was completely different in the hippocampus vs. the plasma. Two-way ANOVA revealed significant interaction effects of HU and genotype in expression levels of five cytokines (out of 35) in socially-housed animals. Elevation of these generally pro-inflammatory cytokines by HU in WT mice was mitigated in mCAT mice, suggesting a role for mitochondrial ROS signaling in inflammatory central nervous system (CNS) responses to microgravity. Interestingly, some of these cytokines in the hippocampus displayed strong correlations to tissue level of 4-hydorxynonenol, a lipid peroxidation product that reflects oxidative damage. We also found substantive cytokine responses to social isolation in the hippocampus; housing and genotype interaction effects were significant (by 2-Factor ANOVA) for 15 cytokines, many of which were mitigated in mCAT mice. Taken together, our results showed that both simulated microgravity and social isolation influenced cytokine levels in the hippocampus and mCAT mice were at least partially protected from these changes. These findings implicate a potentially important role for mitochondrial reactive oxygen species (ROS) in CNS and immune responses to the challenges posed by long duration spaceflight. Reference: Amber M. Paul, Siddhita D. Mhatre, Egle Cekanaviciute, Ann-Sofie Schreurs, Candice G.T. Tahimic, Ruth K. Globus, Brian Crucian, and Sharmila Bhattacharya. Reduced gravity induces oxidative stress and elevates Neutrophil to Lymphocyte ratio in rodents and humans. Submitted May 2019.
Bibliography Type:	Description: (Last Updated: 06/04/2025)
Articles in Peer-reviewed Journals	Tahimic CGT, Paul AM, Schreurs AS, Torres SM, Rubinstein L, Steczina S, Lowe M, Bhattacharya S, Alwood JS, Ronca AE, Globus RK. "Influence of social isolation during prolonged simulated weightlessness by hindlimb unloading." Front Physiol. 2019 Sep 13;10:1147. https://doi.org/10.3389/fphys.2019.01147 ; PMID: 31572207; PMCID: PMC6753329 , Sep-2019